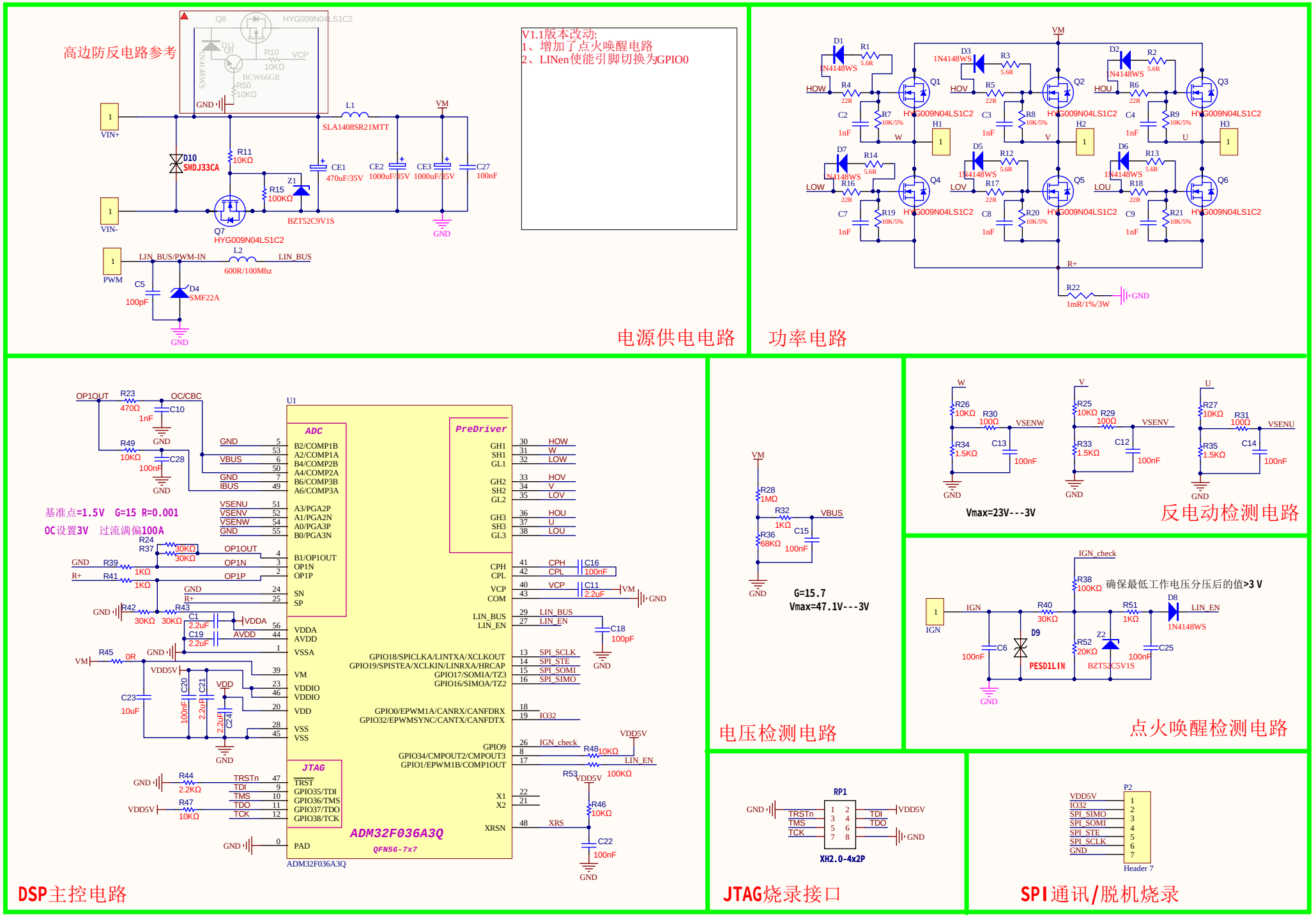

高边防反电路参考
Q8
HYG009N04LS1C2
D11
1N4148WS
R10
10KΩ
VCP
BCW66GR
R50
10KΩ
GND
VIN+
VIN-
D10
SMDJ33CA
R11
10KΩ
R15
100KΩ
Z1
BZT52C9V1S
Q7
HYG009N04LS1C2
L1
SLA1408SR21MTT
CE1
470uF/35V
CE2
1000uF/35V
CE3
1000uF/35V
C27
100nF
VM
PWM
LIN_BUS/PWM-IN
L2
600R/100Mhz
LIN_BUS
C5
100pF
D4
SMF22A
V1.1版本改动:
1、增加了点火唤醒电路
2、LINen使能引脚切换为GPIO0
电源供电电路
D1
D2
D3
1N4148WS
R1
R2
R3
5.6R
HOW
HOV
HOU
R4
R5
R6
22R
Q1
Q2
Q3
C2
C3
C4
1nF
R7
R8
R9
10K/5%
HYG009N04LS1C2
H1
H2
H3
W
V
U
D5
D6
D7
R12
R13
R14
LOW
LOV
LOU
R16
R17
R18
Q4
Q5
Q6
C7
C8
C9
R19
R20
R21
R+
R22
1mR/1%/3W
功率电路
U1
ADC
PreDriver
B2/COMP1B
A2/COMP1A
B4/COMP2B
A4/COMP2A
B6/COMP3B
A6/COMP3A
A3/PGA2P
A1/PGA2N
A0/PGA3P
B0/PGA3N
B1/OP1OUT
OP1N
OP1P
SN
SP
VDDA
AVDD
VSSA
VM
VDDIO
VDD
VSS
PAD
GH1
SH1
GL1
GH2
SH2
GL2
GH3
SH3
GL3
CPH
CPL
VCP
COM
LIN_BUS
LIN_EN
GPIO18/SPICLKA/LINTXA/XCLKOUT
GPIO19/SPISTEA/XCLKIN/LINRXA/HRCAP
GPIO17/SOMIA/TZ3
GPIO16/SIMOA/TZ2
GPIO0/EPWM1A/CANRX/CANFDRX
GPIO32/EPWMSYNC/CANTX/CANFDTX
GPIO9
GPIO34/CMPOUT2/CMPOUT3
GPIO1/EPWM1B/COMP1OUT
X1
X2
XRSN
JTAG
TRST
GPIO35/TDI
GPIO36/TMS
GPIO37/TDO
GPIO38/TCK
ADM32F036A3Q
QFN56-7x7
ADM32F036A3Q
基准点=1.5V G=15 R=0.001
OC设置3V 过流满偏100A
OP1OUT
OC/CBC
VBUS
IBUS
VSENU
VSENV
VSENW
R+
IGN_check
SPI_SCLK
SPI_STE
SPI_SOMI
SPI_SIMO
IO32
XRS
DSP主控电路
R28
1MΩ
R32
1KΩ
R36
68KΩ
C15
100nF
G=15.7
Vmax=47.1V---3V
电压检测电路
R26
R30
R34
R25
R29
R33
R27
R31
R35
C13
C12
C14
VSENW
VSENV
VSENU
Vmax=23V---3V
反电动检测电路
IGN
R38
100KΩ
确保最低工作电压分压后的值>3 V
R40
30KΩ
R51
1KΩ
D8
1N4148WS
LIN_EN
D9
PESD1LIN
Z2
BZT52C5V1S
R52
20KΩ
C6
C25
100nF
点火唤醒检测电路
RP1
TRSTn
TMS
TCK
TDI
TDO
XH2.0-4x2P
JTAG烧录接口
P2
Header 7
SPI通讯/脱机烧录